Deep uncertainty project data and analysis

Raw data

The raw data is in the `/Raw Data` folder. It is made up of individual `.csv` files for each session plus an extra `.csv` file recording page times for all sessions.

The raw data can be cleaned using the `0_clean_raw_data.R` script, that imports all the individual raw data files, cleans, reshapes, and generates `/Clean Data/all_data_evre2.csv`.

Datasets

There are **two** dataset containing clean data, ready to be used.

- `/Clean Data/all_data_evre2.csv` is generated as detailed above and contains all data from the **EVRE2** campaign (choice between safe and risky option.
- `/Clean Data/aggregate_data_evre1.csv` is generated as a byproduct of the **EVRE1** campaign and data analysis, and contains aggregate results for all tasks, questionnaires, and the DRM by subject, for each of 159 subjects taking part in EVRE1; each task and questionnaire measure is the mean over the 8 repetitions; the DRM measures are the mean over 14 days, as seen from the individual and the judges' perspectives separately, and are computed overall and by category.

Variables

EVRE2

Variable	Type	Description
`session`	*text*	unique session identifier
`date`	*date*	date the session was held
`sample`	*text*	nature of the sample, one of "old_subjects" or "new_subjects"
`id`	*text*	8-digit unique oTree identifier for the subject
`label`	*text*	human-readable unique identifier only for `old_subjects`
`period`	*int*	ID andorder of the repetition of the card task [1..17]
`deck`	*text*	unique identifier of the deck of cards used in `period`
`CQtrials`	*int*	number of times subject tried to validate the control questions [1..3]
`CQerrors_t1`	*int*	number of mistakes made in the first CQ trial [0..5]
`CQerrors_t2`	*int*	number of mistakes made in the second CQ trial [0..5]
`CQerrors_total`	*int*	Total number of errors (sum of `CQerrors_t1` and `CQerrors_t2`)
`age`	*int*	Age of the subject, in years
`gender`	*text*	Gender of the subject (male, female, other)
`payoff_variable`	*float*	Variable (incentivized) part of the final payoff
`payoff_fixed`	*float*	Fixed part of the final payoff (15€)
`payoff`	*float*	Total payoff, sum of `payoff_variable` and `payoff_fixed`
`paid_round`	*int*	`period` randomly drawn to be paid
`paid_lottey`	*text*	`deck` used in `paid_round` and hence paid
`paid_step`	*int*	Step paid, 1 (Deep Uncertainty) to 5 (Risk)
`lab_position`	*int*	Unique identifier of the cubicle the subjects sat at
`timespent_seconds`	*int*	Time spent for the whole experiment, from the start to the end of the main task
`choice_du_pre`	*dummy*	Choice in the deep uncertainty step 1. 0 = safe, 1 = risky
`choice_du_post`	*dummy*	Choice in the deep uncertainty + draw step 2. 0 = safe, 1 = risky
`choice_amb_pre`	*dummy*	Choice in the ambiguity step 3. 0 = safe, 1 = risky
`choice_amb_post`	*dummy*	Choice in the ambiguity + draw step 4. 0 = safe, 1 = risky
`choice_risky`	*dummy*	Choice in the risk step 5. 0 = safe, 1 = risky
`perception_1 .. 5`	*int*	Risk perception in step 1 .. 5 [0..10]

Variable	Type	Description
`draw1 .. draw12`	*float*	first .. twelfth card drawn by the subject
`v1 .. v4`	*float*	values of the `deck` used in that round
`p1 .. p4`	*float*	probabilities associated with `v1 .. v4`
`EV`	*int*	expected value of the `deck` (always 20)
`n.outcomes`	*int*	number of different values in the `deck` [2..4]
`losses`	*dummy*	did the deck contain losses?
`variance`	*float*	variance of the `deck`
`prob`	*text*	type of probability distribution of `deck` ["equal", "leaning bad", "leaning good", "unequal sym"]
`name`	*text*	human-readable name describing `deck`

EVRE1

Variable	Type	Description
`label`	*text*	human-readable unique identifier
`BRET_choice`	*float*	Mean choice in the BRET over 8 repetitions [0..100]
`BRET_perception_decision`	*float*	Mean risk perception of the choice in the BRET over 8 repetitions [0..10]
`DOSPERT`	*float*	Mean DOSPERT overall score over 8 repetitions
`DOSPERT_eth`	*float*	Mean DOSPERT ethical score over 8 repetitions
`DOSPERT_fin`	*float*	Mean DOSPERT financial score over 8 repetitions
`DOSPERT_hlt`	*float*	Mean DOSPERT health score over 8 repetitions
`DOSPERT_rec`	*float*	Mean DOSPERT recreational score over 8 repetitions
`DOSPERT_soc`	*float*	Mean DOSPERT social score over 8 repetitions
`HL_numrisky`	*float*	Mean number of risky choices in HL over 8 repetitions [0..10]
`HL_perception_decision`	*float*	Mean risk perception of the choice in the HL over 8 repetitions [0..10]
`IG_choice`	*float*	Mean amount invested in IG over 8 repetitions [0..4]
`IG_perception_decision`	*float*	Mean risk perception of the choice in the IG over 8 repetitions [0..10]
`LOSS_numrisky`	*float*	Mean number of risky choices in LOSS over 8 repetitions [0..7]
`LOSS_perception_decision`	*float*	Mean risk perception of the choice in LOSS over 8 repetitions [0..10]
`soep`	*float*	Mean SOEP overall score over 8 repetitions
`soep_driving`	*float*	Mean SOEP driving score over 8 repetitions
`soep_financial`	*float*	Mean SOEP financial score over 8 repetitions
`soep_recreational`	*float*	Mean SOEP recreational score over 8 repetitions
`soep_trust`	*float*	Mean SOEP trust score over 8 repetitions
`soep_work`	*float*	Mean SOEP work score over 8 repetitions
`Avoid_N`	*int*	DRM: number of activities at risk where subject avoided risk
`Take_N`	*int*	DRM: number of activities at risk where subject took risk
`Take_share`	*float*	DRM: share of `Take_N` in total number activities over 14 days
`Avoid_share`	*float*	DRM: share of `Avoid_N` in total number activities over 14 days
`Weighted_risk`	*float*	DRM: mean self-declared risk taken over 14 days
`BRET_r`	*float*	Risk aversion parameter r implied by the BRET choices
`HL_r`	*float*	Risk aversion parameter r implied by the HL choices
`IG_r`	*float*	Risk aversion parameter r implied by the IG choices
`LOSS_lambda`	*float*	Loss aversion parameter λ implied by the LOSS choices
`DRM_all_activities`	*float*	DRM: mean self-declared risk taken over 14 days [identical to `Weighted_risk`)
`DRM_unanimity`	*float*	DRM: mean self-declared risk taken over 14 days, excluding activities where *all judges* thought the activity contained actually no risk at all

Variable	Type	Description
`DRM_majority`	*float*	DRM: mean self-declared risk taken over 14 days, excluding activities where *at least 3 out of 4 judges* thought the activity contained actually no risk at all
`DRM_judge_1`	*float*	DRM: mean level of risk over 14 days, as rated by judge 1
`DRM_judge_2`	*float*	DRM: mean level of risk over 14 days, as rated by judge 2
`DRM_judge_3`	*float*	DRM: mean level of risk over 14 days, as rated by judge 3
`DRM_judge_4`	*float*	DRM: mean level of risk over 14 days, as rated by judge 4